A Positive Daily Return in the S&P 500 (SPX) Increases the Probability of a Positive Return in the Same Direction for the NASDAQ Composite (NAS) on the Following Trading Day, More Than a Random Chance Would Suggest.

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Hypothesis

The U.S. stock market is one of the largest financial systems in the world. It includes 3,558 publicly traded companies with a combined market capitalization of around $70.976 trillion, which underlines its scale and global importance (CompaniesMarketCap, 2025). Because it attracts global capital and reacts quickly to new information, it is a useful setting for studying short-run market dynamics and interactions between major benchmarks.

Stock indices often move together because investors react to the same headlines, macroeconomic signals, and shifts in risk appetite (Investopedia, 2025). The S&P 500 (SPX) is widely used as a benchmark for U.S. equity market performance and tracks 500 leading firms (S&P Dow Jones Indices, 2025). The NASDAQ Composite (NAS) measures the performance of more than 2,500 companies listed on the Nasdaq Stock Market and is more heavily weighted toward technology and growth-oriented firms (Nasdaq, 2025). Since these indices overlap in investor attention but differ in composition, it is reasonable to expect short-term spillovers where one index may partially lead another.

In this study, we examine whether there is a short-term relationship between the S&P 500 and the NASDAQ Composite with a one-trading-day lag. The hypothesis states that a positive daily return in the S&P 500 increases the probability that the NASDAQ will generate a return in the same direction on the following trading day, more than would be expected by pure chance.

To test this hypothesis, we apply a time-series regression model using daily data. The dependent variable is tomorrow's NASDAQ return, constructed by shifting NASDAQ returns forward by one trading day. Since U.S. stock exchanges operate on business days and close on weekends and holidays, the sample includes only trading sessions to avoid artificial breaks and to ensure lagged variables are defined correctly (NYSE, 2025). The main explanatory variable is the current daily return of the S&P 500. The model also includes additional control variables capturing market trend, short-term risk, and monthly seasonality in order to account for broader conditions that may affect return dynamics.

Data Commentary

The selected data set for analysis is a financial time series dataset from Kaggle, covering the period from January 2023 to December 2025 and containing daily market data on U.S. indices. Initially, the dataset contained a wide range of market indicators, but for the current analysis, we focused on a specific subset of variables relevant to modeling market dynamics and predicting the transformed NASDAQ return, cleaning data with the Pandas module in Python.

Table 1

Data Commentary of dependent and independent variables

Variable	**Explanation**
NASDAQ_NextDay_Return_ Transformed (nq_transf)	Dependent numeric variable. This variable is a converted version of the NASDAQ return for the next day, calculated (in Gretl) as: series nq_transf = NASDAQ_Return_1 > 0 ? log(1 + NASDAQ_Return_1) : (NASDAQ_Return_1 < 0 ? -log(1 - NASDAQ_Return_1) : 0); where “NASDAQ_Return_1” is a series of NASDAQ_Return with 1 lag, meaning is tomorrow's NASDAQ return. During the conversion, a logarithmic scale is applied to the value of profitability while maintaining its sign (plus or minus). This makes it possible to normalize the distribution of returns and reduce the excessive impact of extreme deviations, making it a more stable target variable for predictive modeling.
time	Independent time trend variable. The market launch date for the registered observations. This variable is important for maintaining the order of data in time series, conducting time analysis, and for developing potential characteristics (for example, to determine seasonal effects or periods of specific events). It is not used in models as a direct predictor.
SPX_Return	Independent numeric variable. Daily percentage return of the S&P 500 index. This variable reflects the dynamics of the US stock market as a whole and is used as a key

	characteristic to determine the overall mood and direction of market movement on a given day.
Market_Regime	Independent categorical variable. Classification of dominant market conditions for a given day, classified as "Bull", "Neutral" or "Bear". This variable is crucial for contextualizing other indicators, as market behavior, correlations, and model effectiveness can vary significantly depending on these modes.
Deviation_from_SMA	Independent numeric variable. The percentage difference between the current closing price of the S&P 500 and its 50-day simple moving average (SMA). A positive value indicates that the index is trading above its medium-term trend (potential overbought), while a negative value indicates that it is lower (potential oversold). It is used as an indicator of dynamics and returns to the average value.
SPX_Volatility_10D	Independent numeric variable. A measure of the volatility of S&P 500 returns over the past 10 days (usually a moving standard deviation). This variable quantifies the current market uncertainty and risk environment. Higher volatility often coincides with market tensions and can affect the magnitude and predictability of returns.

Descriptive Statistics

The frequency distribution analysis performed for the independent categorical variable "Market_Regime" shows that the three market condition categories are not equally frequent, as detailed in the table below:

Table 2

Frequency distribution for the independent variable "Market_Regime"

Market Regime	**Frequency**	**Percentage**
Bull	352	48.55%
Neutral	289	39.86%
Bear	84	11.59%
Total	725	100%

The observed proportions reflect market conditions from January 2023 to December 2025. The dataset shows that the “bullish” mode was the most common (48.55%), which is typical for almost half of the trading days. It is followed by a “neutral” regime (39.86%). Together, these two categories of stable or positive moods account for 88.41% of the entire period. The "bearish" regime is the least common (11.59%), which indicates that pronounced downward trends were relatively rare. This distribution is largely consistent with the overall upward trajectory and recovery of the major U.S. markets seen during this sample period.

Table 3

Summary statistics of key continuous independent and dependent variables

Variable	Mean	Median	Min	Max	Range	Skewness
SPX_Return (%)	0.0812	0.073	-5.09	10.54	15.63	1.136
nq_transf (%)	0.076	0.1	-1.81	2.63	4.44	-0.163
Deviation_from_SMA (%)	1.3	1.93	-13.71	7.87	21.58	-1.235
SPX_Volatility_10D (%)	0.695	0.616	0.083	4.67	4.59	5.075

For the SPX_Return, the mean (0.081%) is slightly higher than the median (0.073%). This difference, combined with a significant positive skewness of 1.136, indicates a strong rightward shift in the distribution. This is a significant change from the analysis and suggests that during this period, large positive daily returns were more frequent or significant than large negative ones. The maximum return of 10.54% is significantly higher than the minimum of -5.09%, which confirms this pattern.

The transformed NASDAQ index, nq_transf, shows a mean (0.076%), which is lower than its median (0.1%). The skewness of -0.163 is very close to zero and negative, indicating that the logarithmic transformation was very effective in normalizing the distribution. The resulting series is much more symmetrical than the original one, with a range of values strictly limited from -1.81 to 2.63.

The variable Deviation_from_SMA has a mean (1.30%) which is significantly lower than its median (1.93%). The pronounced negative skewness of -1.235 confirms that the distribution is strongly shifted to the left. This pattern corresponds to a market that has been trading above its 50-day moving average most of the time (hence the positive median), but has experienced several sharp, large deviations below it (as seen at the low of -13.71%), which has led to a decrease in the mean.

Most strikingly, SPX_Volatility_10D shows an extreme positive skewness of 5.075. The average volatility (0.695%) is significantly higher than the median (0.616%) due to the very long right tail, which reaches a maximum value of 4.67%. This confirms the extremely intense "clustering of volatility" characteristic of financial markets: while most observations reflect calm conditions (clustering is close to the average), rare periods of severe market stress have led to spikes in volatility of extraordinary magnitude.

Stationarity Testing

Before running the regression, we first checked whether the two main return series, SPX_Return and NASDAQ_Return, are stationary. This matters because if a series has a unit root, it can create fake relationships in a model that look meaningful only because both variables drift over time.

Figure 1

Results of the Augmented Dickey-Fuller (ADF) Test for Stationarity of SPX Returns

gretl: ADF test

```
Augmented Dickey-Fuller test for SPX_Return
testing down from 17 lags, criterion AIC
sample size 721
unit-root null hypothesis: a = 1

  test with constant
  including 3 lags of (1-L)SPX_Return
  model: (1-L)y = b0 + (a-1)*y(-1) + ... + e
  estimated value of (a - 1): -1.17373
  test statistic: tau_c(1) = -14.7228
  asymptotic p-value 2.479e-34
  1st-order autocorrelation coeff. for e: -0.002
  lagged differences: F(3, 716) = 1.325 [0.2651]

  with constant and trend
  including 3 lags of (1-L)SPX_Return
  model: (1-L)y = b0 + b1*t + (a-1)*y(-1) + ... + e
  estimated value of (a - 1): -1.17373
  test statistic: tau_ct(1) = -14.7124
  asymptotic p-value 1.51e-41
  1st-order autocorrelation coeff. for e: -0.002
  lagged differences: F(3, 715) = 1.323 [0.2658]
```

Figure 2

Results of the KPSS Test for Stationarity of SPX Returns

gretl: KPSS test

```
KPSS test for SPX_Return

T = 725
Lag truncation parameter = 17
Test statistic = 0.0346354

                    10%      5%      1%
Critical values: 0.348   0.462   0.743
P-value > .10
```

Starting with SPX_Return, the ADF test was run with lag selection based on AIC (testing down from 17 lags), and the final model included 3 lagged differences. The output

shows a test statistic of −14.7228 with a p-value of 2.479e−34 under the constant specification, which clearly rejects the unit-root null. Adding a trend does not change the conclusion: t = −14.7124 with p = 1.51e−41. The KPSS test then confirms the same story. Its statistic is 0.0346354, far below the critical values, and the p-value is above 0.10, so we do not reject stationarity.

Figure 3

Results of the Augmented Dickey-Fuller (ADF) Test for Stationarity of NASDAQ Returns

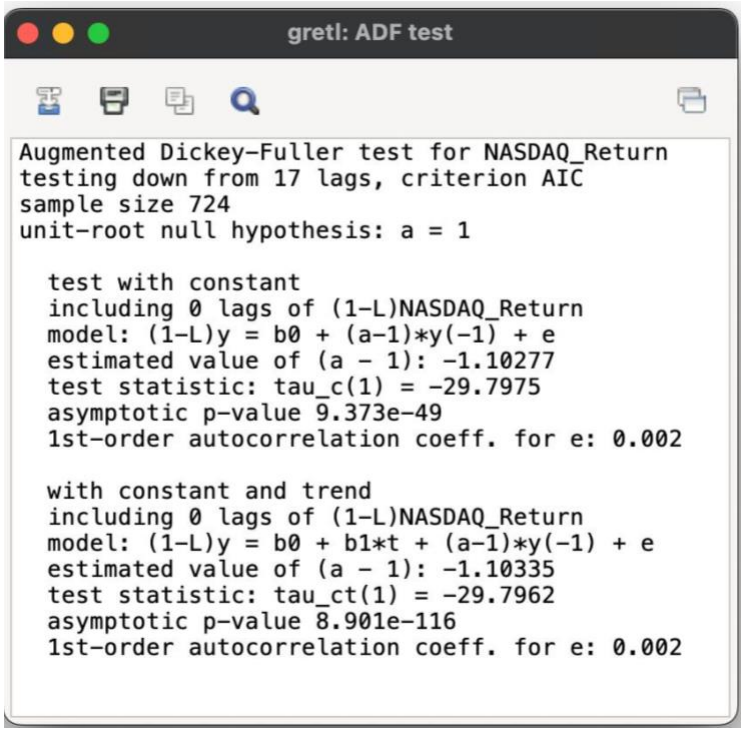
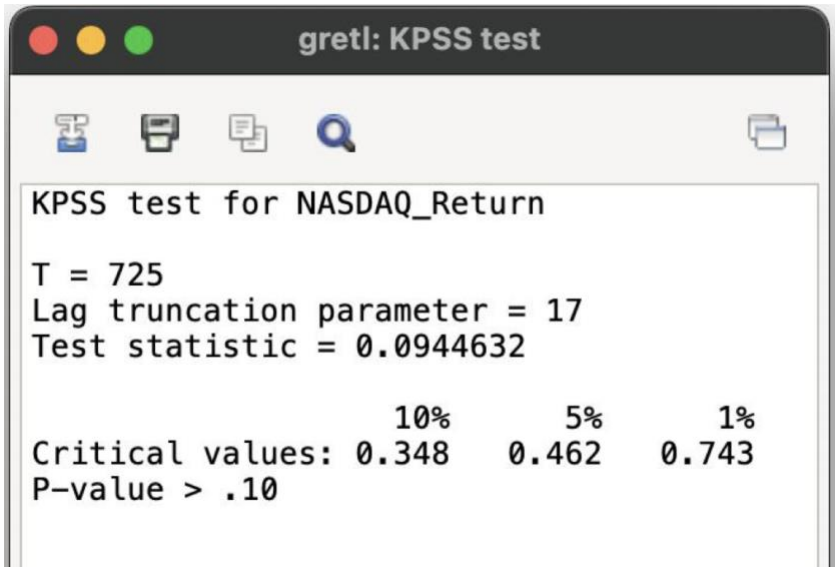


Figure 4

Results of the KPSS Test for Stationarity of NASDAQ Returns



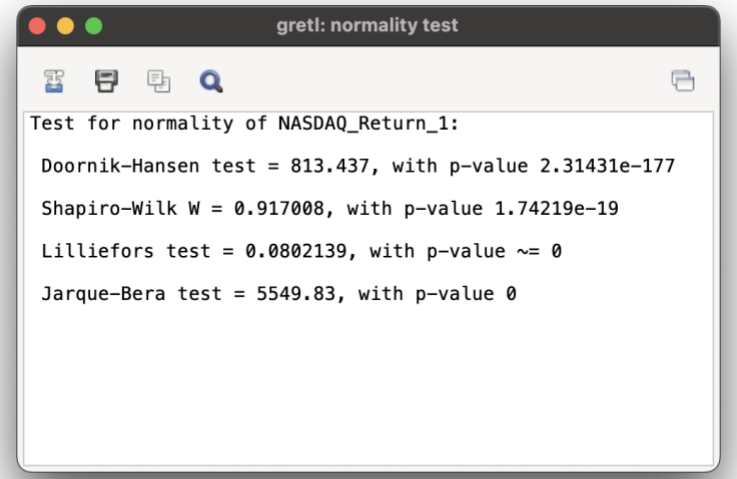
The same pattern appears for NASDAQ_Return. The ADF test is extremely strong here, with t = −29.7975 (p = 9.373e−49) with a constant and τ = −29.7962 (p = 8.901e−116) with constant and trend, leaving no doubt that the series is stationary. The KPSS statistic is 0.0944632, again well below all critical values, with p-value > 0.10. So, both SPX_Return and NASDAQ_Return are stationary in levels. That means we can keep the data in its original return form, without differencing, and move forward knowing the regression is testing a real short-run relationship rather than an artificial pattern in unstable time-series data.

Normality of Dependent Variable

Originally, the dependent variable used to be NASDAQ_Return_1 variable (NASDAQ_Return with 1 lag). However, after checking normality tests of this variable, it was found that none of those tests have conducted normality of the variable: all statistical tests except Doornik-Hansen and Shapiro-Wilk W (with p-values 2.314131e-177 and 1.74219e-19, respectively) yield p-values of 0 or approaching 0, consistently indicating a strong violation of normality.

Figure 5

Results of Normality Tests for NASDAQ_Returns_1



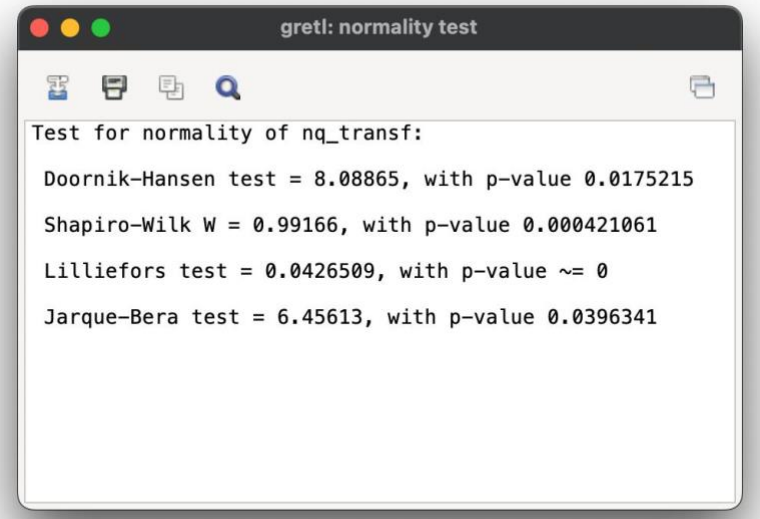
To address this, we applied a logarithmic transformation:

series nq_transf = NASDAQ_Return_1 > 0 ? log(1 + NASDAQ_Return_1) : (NASDAQ_Return_1 < 0 ? -log(1 - NASDAQ_Return_1) : 0).

After that, Doornik-Hansen, Shapiro-Wilk W, Lilliefors, and Jarque-Bera tests were conducted for our new dependent variable "nq_transf", as well as visual diagnostics including a Q-Q plot (Figure 7) and kernel density estimation (Figure 8). The results from the statistical tests present a mixed but generally significant picture.

Figure 6

Results of Normality Tests for Transformed NASDAQ Returns (nq_transf)



Using a significance level of α = 0.05, three of the four tests (Doornik-Hansen, Shapiro-Wilk, and Lilliefors) yield p-values below 0.05, indicating a rejection of the null hypothesis of normality. The Jarque-Bera test shows a p-value just below the threshold, exceeding a significance level of 0.01. Collectively, this suggests a violation of the normality assumption for the nq_transf variable, but still significantly better than original NASDAQ_Return_1.

Figure 7

Q-Q plot for the "nq_transf" variable

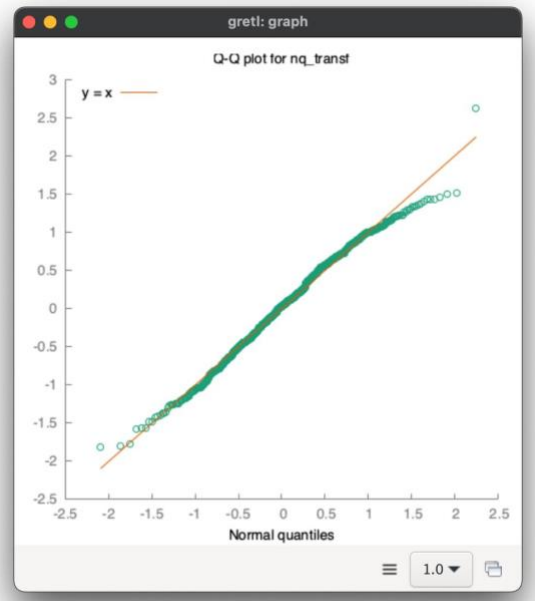
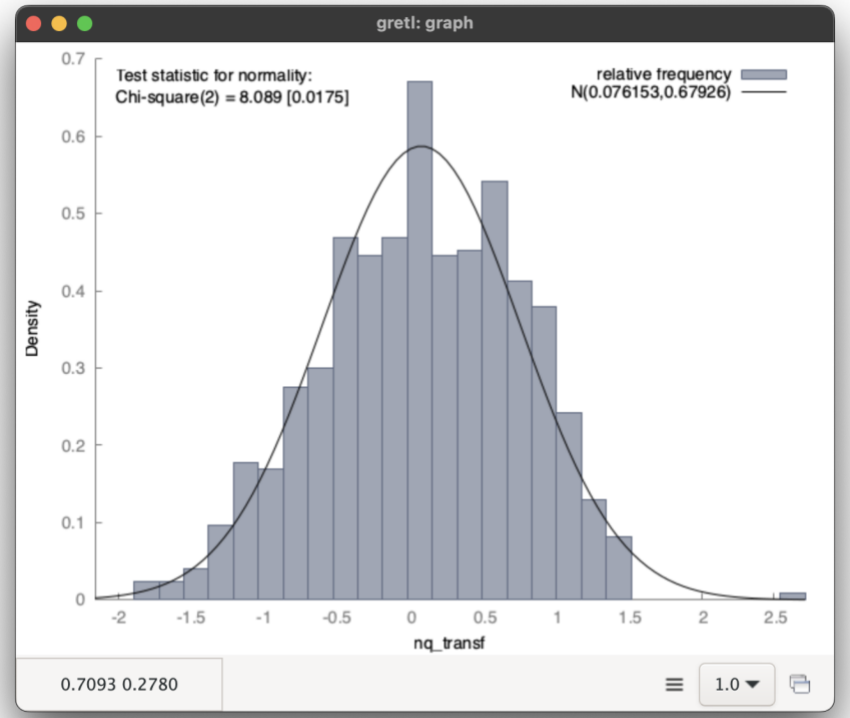


Figure 8

Kernel density plot for the "nq_transf" variable



The Q-Q plot shows deviations from the diagonal reference line in the tails, indicating heavier tails than a normal distribution. The kernel density plot reveals a distribution that is roughly symmetric but with excess kurtosis (heavier tails and a sharper peak) compared to a normal distribution, which is typical for financial return data even after transformation.

This logarithmic transformation was applied to reduce the effect of extreme values and stabilize the variance. The persistent non-normality revealed during the tests indicates that although the transformation improved the properties of the distribution, it did not lead to the achievement of ideal normality. This is a common result in financial time series, where residual and volatility clustering often remain. For the purposes of Ordinary Least Squares (OLS) regression, the central limit theorem often serves as a justification for the conclusion, since the asymptotic normality of the estimate can be maintained even if the dependent variable itself is not distributed quite normally. Therefore, the current transformed variable nq_transf is considered suitable for analysis confirming the noted deviations from the norm.

OLS Regression Model

To evaluate the predictive relationship between the returns of the S&P 500 and the NASDAQ Composite, we estimated an OLS regression model. The dependent variable was the transformed NASDAQ variable (nq_transf).

Figure 9

Results of OLS Regression for Transformed NASDAQ Returns (nq_transf)

gretl: model 1

File Edit Tests Save Graphs Analysis LaTeX

```
Model 1: OLS, using observations 2023-01-20:2025-12-04 (T = 724)
Dependent variable: nq_transf

                      coefficient  std. error  t-ratio   p-value
  ---------------------------------------------------------------
  const               -0.252677    0.0935091   -2.702    0.0071   ***
  SPX_Return          -0.0725826   0.0601320   -1.207    0.2278
  Market_Regime        0.0264609   0.0375684    0.7043   0.4815
  Deviation_from_S~    0.0904021   0.0112485    8.037    3.82e-15 ***
  SPX_Volatility_1~    0.224627    0.0640147    3.509    0.0005   ***
  dummy_2              0.193823    0.0774082    2.504    0.0125   **
  dummy_3             -0.0437472   0.0776177   -0.5636   0.5732
  dummy_4              0.00227170  0.0776231    0.02927  0.9767
  dummy_5             -0.0531862   0.0775253   -0.6860   0.4929
  SPX_interact_term   -0.0665920   0.0651310   -1.022    0.3069

Mean dependent var   0.076153   S.D. dependent var   0.679260
Sum squared resid    297.8802   S.E. of regression   0.645910
R-squared            0.107041   Adjusted R-squared   0.095785
F(9, 714)            9.509848   P-value(F)           9.22e-14
Log-likelihood      -705.8192   Akaike criterion     1431.638
Schwarz criterion    1477.486   Hannan-Quinn         1449.334
rho                  0.001400   Durbin-Watson        1.995862

Excluding the constant, p-value was highest for variable 29 (dummy_4)

LM test for autocorrelation up to order 6 -
  Null hypothesis: no autocorrelation
  Test statistic: LMF = 0.343859
  with p-value = P(F(6, 709) > 0.343859) = 0.913533
```

The model includes continuous variables reflecting market returns for the same day and conditions such as the return of the S&P 500 index (SPX_Return), its deviation from trend (Deviation_from_SMA) and recent market volatility (SPX_Volatility_10D). To account for potential changes in market dynamics, categorical variables for structural shifts were included using dummy variables.

We used dummy variables to account for discrete, one-time changes in the data generation process at certain points in time, since these structural shifts are qualitative events without a natural numerical ordering. In total, four dummy variables and one interaction element were included.

In order to calculate SPX_interact_term (to account for a potential shift in the return relationship), a structural break dummy variable was created in Q2 2024:

d_break = 1 for observations on or after April 9, 2024, and 0 otherwise.

This dummy variable is included both as an intercept shifter and in an interaction term: SPX_interact_term (d_break * SPX_Return). This setup allows the model to estimate whether the baseline next-day NASDAQ return (intercept) and, crucially, the sensitivity to SPX returns (slope coefficient) changed significantly after the break date.

dummy_2, dummy_3, dummy_4, dummy_5: Binary variables equal to 1 for observations occurring after four different days of a week (from Tuesday to Friday; Monday was excluded as a reference dummy variable (dummy_1), and Saturday and Sunday were excluded since there are no regular trading hours for indexes on these days of a week), and 0 otherwise.

SPX_interact_term: An interaction between SPX_Return and one of the significant dummy interrupt elements (implicitly dummy_2), verifying a change in predictive dependency after this interrupt.

This setup allows the model to evaluate how market indicators of the same day predict NASDAQ returns for the next day, while monitoring whether this predictive relationship has changed after significant market events compared to the base period.

Normality of Residuals and Heteroscedasticity of Residuals

```
Test for normality of residual -
  Null hypothesis: error is normally distributed
  Test statistic: Chi-square(2) = 6.72349
  with p-value = 0.0346746

White's test for heteroskedasticity -
  Null hypothesis: heteroskedasticity not present
  Test statistic: LM = 228.824
  with p-value = P(Chi-square(43) > 228.824) = 3.55114e-27

Breusch-Pagan test for heteroskedasticity -
  Null hypothesis: heteroskedasticity not present
  Test statistic: LM = 117.733
  with p-value = P(Chi-square(9) > 117.733) = 3.88813e-21
```

At a 5% significance level (α=0.05), the p-value is less than 0.05, but exceeds 0.01 threshold. Still, we reject the null hypothesis. This indicates that the residuals are not normally distributed.

The model also suffers from a violation of the assumptions of heteroscedasticity. It is a particularly serious problem because it leads to inefficient estimates and biased standard errors affecting hypothesis testing.

Multicollinearity

```
Variance Inflation Factors
Minimum possible value = 1.0
Values > 10.0 may indicate a collinearity problem

          SPX_Return    5.990
       Market_Regime    1.063
  Deviation_from_SMA    1.514
  SPX_Volatility_10D    1.458
             dummy_2    1.683
             dummy_3    1.692
             dummy_4    1.705
             dummy_5    1.688
SPX_interaction_term    5.827

VIF(j) = 1/(1 - R(j)^2), where R(j) is the multiple correlation coefficient
between variable j and the other independent variables
```

```
Belsley-Kuh-Welsch collinearity diagnostics:

  variance proportions

    lambda      cond     const SPX_Retu~ Market_R~ Deviatio~ SPX_Vola~   dummy_2   dummy_3
     3.776     1.000     0.004     0.001     0.007     0.009     0.011     0.007     0.007
     1.967     1.386     0.000     0.038     0.001     0.014     0.001     0.002     0.000
     1.004     1.940     0.000     0.000     0.000     0.007     0.001     0.298     0.000
     1.000     1.943     0.000     0.000     0.000     0.000     0.000     0.047     0.121
     0.993     1.950     0.000     0.001     0.000     0.004     0.000     0.009     0.234
     0.818     2.148     0.000     0.006     0.001     0.487     0.038     0.006     0.002
     0.170     4.713     0.000     0.000     0.072     0.135     0.199     0.440     0.463
     0.137     5.242     0.016     0.003     0.383     0.203     0.649     0.025     0.015
     0.088     6.562     0.003     0.950     0.003     0.032     0.030     0.002     0.005
     0.047     8.925     0.976     0.002     0.534     0.109     0.071     0.166     0.152

    lambda      cond   dummy_4   dummy_5 SPX_inte~
     3.776     1.000     0.007     0.007     0.001
     1.967     1.386     0.003     0.000     0.038
     1.004     1.940     0.030     0.134     0.000
     1.000     1.943     0.097     0.207     0.000
     0.993     1.950     0.216     0.009     0.000
     0.818     2.148     0.000     0.005     0.009
     0.170     4.713     0.502     0.479     0.010
     0.137     5.242     0.015     0.013     0.022
     0.088     6.562     0.000     0.000     0.914
     0.047     8.925     0.129     0.145     0.006

  lambda = eigenvalues of inverse covariance matrix (smallest is 0.0473968)
  cond   = condition index
  note: variance proportions columns sum to 1.0

According to BKW, cond >= 30 indicates "strong" near linear dependence,
and cond between 10 and 30 "moderately strong".  Parameter estimates whose
variance is mostly associated with problematic cond values may themselves
be considered problematic.

Count of condition indices >= 30: 0
Count of condition indices >= 10: 0

No evidence of excessive collinearity
```

No single variable demonstrates multicollinearity in accordance with the VIF criterion. Increased VIF values for SPX_Return and its interaction conditions are expected, since

interaction conditions often correlate with their components, but with a value of 5.99, this is not a concern.

Autocorrelation of Residuals

Both tests, Durbin-Watson and LM Test, confirm that the residual values do not show significant patterns of autocorrelation, since DW test's value is very close to 2 and LM test's p-value is extremely high (0.91 > 0.05).

Granger Causality Testing

Granger causality tests were used in order to examine the predictive relationships for SPX_Return and nq_transf. Both variables were treated as endogenous and their dynamic interdependency was taken into account in the model. Since both time series appeared to be stationary, as shown in Figure 1 and 3, the classical Granger causality test is asymptotically correct. In a given case, the Toda-Yamamoto method gives the same results, as the maximal order of integration is zero, and no additional lags are required for a correlated Wald test.

As previously shown in Figure 9, the results suggest a statistically significant Granger causality implying there is an indication of past values on one variable providing useful information for predicting the other.

However, these findings should not be misinterpreted to suggest a pattern of structural or economic causality. In financial markets, profit return series that are such as the SPX_Return and the transformed Nasdaq return (nq_transf) are by their nature prone to endogeneity, being simultaneously responsive to common information shocks, which include macroeconomic news, monetary policy expectations, and changes in global risk-averse biases.

This concurrence results in what is often described as pseudo-causality, where Granger causality reflects information flows or synchronised market adjustments rather than a true causal transmission mechanism. As a result, the observed Granger causality could be the result of some overlapping sets of information and feedback effects, rather than a unidirectional effect from one market to another.

Overall, the results of Granger causality tests are best interpreted as evidence of dynamic predictability under endogeneity, in keeping consistent with the concept that stock market returns evolve together by responding to common economic signals. While the VAR model takes into account feedback dynamics, it does not remove endogeneity, and thus these results are meant to be considered more as descriptive rather than structural.

Interpretation of Regression Results

The following model was used to evaluate the predictive relationship between S&P 500 and NASDAQ Composite returns:

$nq_transf_t = \beta_0 + \beta_1(SPX_Return_t) + \beta_2(Market_Regime_t) + \beta_3(Deviation_from_SMA_t) + \beta_4(SPX_Volatility_10D_t) + \beta_5(dummy_2) + \beta_6(dummy_3) + \beta_7(dummy_4) + \beta_8(dummy_5) + \beta_9(SPX_interact_term) + \varepsilon_t;$

Where:

- nq_transf_t is the transformed next-day NASDAQ return for period “t”;
- SPX_Return_t is the daily percentage return of the S&P 500 index;
- Market_Regime_t: A single variable representing the market condition (likely numerical: 1, 2, 3 for Bear, Neutral, Bull).
- Deviation_from_SMA_t is the percentage deviation of the SPX from its 50-day simple moving average;
- SPX_Volatility_10D_t is the 10-day historical volatility of the SPX;
- dummy_2, dummy_3, dummy_4, dummy_5: Four separate dummy variables you created to test for structural breaks at different points in time.
- SPX_interact_term_t is an interaction term capturing a change in the SPX coefficient after the break;
- ε_t is the error term.

This model evaluates how SPX returns on the same day and key market controls predict the direction and magnitude of NASDAQ returns the next day. The transformation of the dependent variable was applied to correct the abnormality. Although it is not a purely logarithmic linear model, the coefficients for continuous variables can be interpreted as a change in the transformed NASDAQ returns when the predictor is changed by one.

SPX_Return Effect

Coefficient: -0.0726 (p-value = 0.2278)

The coefficient for the main variable of the hypothesis is negative and statistically insignificant (p-value > 0.05). This means that after taking other factors into account, there is no reliable statistical evidence in the model that SPX returns on the same day have a linear predictive effect on the transformed NASDAQ returns on the next day. This conclusion does not confirm the main hypothesis of the study.

Deviation_from_SMA Effect

Coefficient: 0.0904 (p-value < 0.001)

A very significant positive effect. A 1 percentage point increase in how far the SPX is trading above its 50-day trend is due to a 0.09 unit increase in NASDAQ's projected return

the following day. This suggests that market dynamics, rather than same-day returns, are a stronger predictor.

SPX_Volatility_10D Effect

Coefficient: 0.2246 (p-value = 0.0005)

A very significant positive effect. Contrary to the expected deterrent effect, the recent 1 percentage point increase in market volatility is due to an increase in NASDAQ's projected profitability by 0.22 units. This may indicate that unstable periods are followed by compensatory bounces or reflect certain dynamics during the period of your sample.

Dummy_2 Effect

Coefficient: 0.1938 (p-value = 0.0125)

A significant positive coefficient for this fictitious gap indicator indicates that the baseline (intersection point) of the converted NASDAQ returns has shifted upward since the period it represents, regardless of other variables.

SPX_interact_term Effect

Coefficient: -0.0666 (p-value = 0.3069)

The interaction term is insignificant. This shows that although the overall level of NASDAQ profitability has changed since the break (dummy_2), the predictive relationship (slope) between SPX_Return and nq_transf has not changed statistically.

The Market_Regime variable and other dummy values (dummy_3, dummy_4, dummy_5) are statistically insignificant (p-values > 0.05). This suggests that in this model specification, these categorical differences do not provide an additional explanation for NASDAQ's next-day returns.

Goodness of fit

The explanatory power of the model is low, the Adjusted R-squared value is 0.095785. This indicates that approximately 90% of the NASDAQ's next-day yield changes are due to factors not included in this model.

The Durbin-Watson test (~1,996) and the LM autocorrelation test (p-value = 0.913) clearly indicate no serial correlation in the residuals, which is a positive sign of the validity of the test statistics.

The main result is that SPX_Return is not a significant predictor in this model. The significant drivers determining this model are Deviation_from_SMA (momentum) and SPX_Volatility_10D, as well as the structural shift in the baseline recorded by dummy_2.

Conclusion

In this paper, we tested the hypothesis that a positive daily return in the S&P 500 increases the chance that the NASDAQ Composite will move in the same direction on the next trading day. In short, we wanted to see whether the S&P 500 contains any real next-day signal for the NASDAQ, beyond what random movement would produce.

To do this, we built a time-series regression model using daily data from January 2023 to December 2025. The dependent variable was the NASDAQ return on the following trading day. Since daily returns can include sharp jumps and extreme values, we used a transformed version of the next-day return (nq_transf) to make the series more stable. We also checked stationarity for both SPX_Return and NASDAQ_Return using the ADF and KPSS tests. Both variables were stationary, so we did not need to difference the data.

The key finding that the hypothesis wasn't supported. The coefficient on SPX_Return was negative (−0.0726) and statistically insignificant ($p = 0.2278$). Put simply, once we control for other factors, the S&P 500 return today does not reliably predict the NASDAQ return tomorrow in a linear way.

We also tested whether this relationship changed after a specific break date by adding a structural dummy and an interaction term. The interaction term remained insignificant ($p = 0.3069$). This suggests that even if average returns shifted over time, the strength of the SPX-to-NASDAQ link did not meaningfully change in the data.

At the same time, the model shows that other variables carry more information. The strongest predictors were Deviation_from_SMA ($p < 0.001$) and SPX_Volatility_10D ($p = 0.0005$). In practice, this means the NASDAQ's next-day return is more connected to the overall market trend and short-term risk environment than to the S&P 500's daily return alone.

The model's explanatory power is low but expected for this type of problem. The adjusted R^2 is 0.0958, so the model explains about 9.6% of the variation in next-day NASDAQ returns. That is typical for daily return forecasting. However, it is important because much of the movement remains outside the pattern and depends on news, macroeconomic surprises, central bank signals, company earnings, and global events in general.

There are two main limitations we want to highlight. First, the sample covers only 2023–2025. This period includes unstable market conditions, frequent changes in interest rate expectations, and strong effects from political events and conflicts, so the results may not look the same in other periods. Second, even after transforming nq_transf, the

distribution is still not fully normal. That is common for financial returns and stocks in general, but it can affect the precision of standard errors and some test statistics.

Overall, in this specification, SPX_Return is not a stable next-day predictor for NASDAQ direction. Market trend and uncertainty are much more revealing.

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Note: The authors used DeepL Translate for selected draft passages that were translated from Russian to English and subsequently reviewed and edited all translated content for accuracy, terminology, and academic tone before including it in the final manuscript.

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Note: "Clarifications of the proper interpretations of the normality tests, kernel density plots, Q-Q plots, and regression tables of the performed work in Gretl were supported by OpenAI's ChatGPT (GPT-4, custom econometrics setup). All AI-generated content was carefully reviewed, edited and validated by the authors."